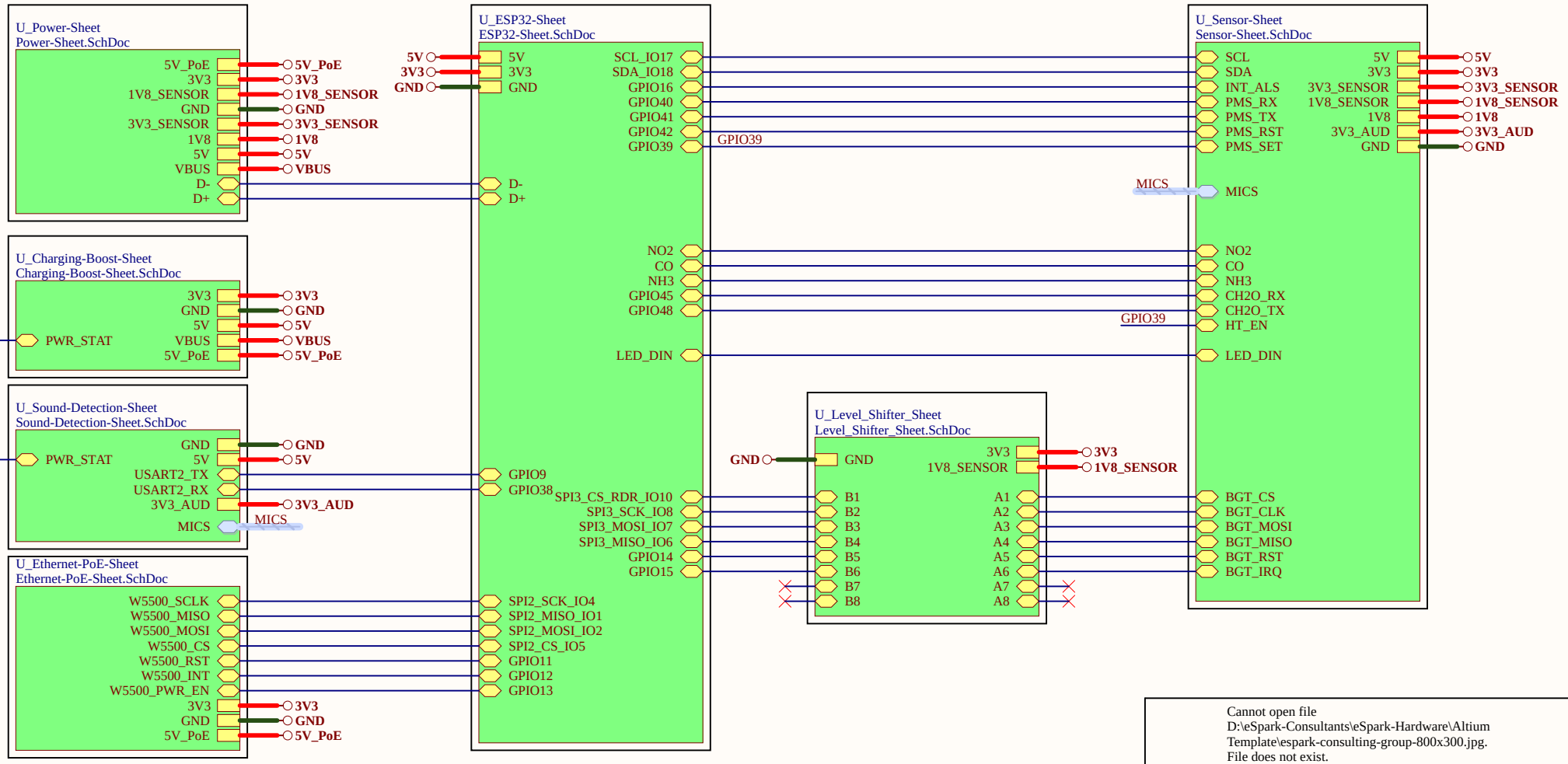
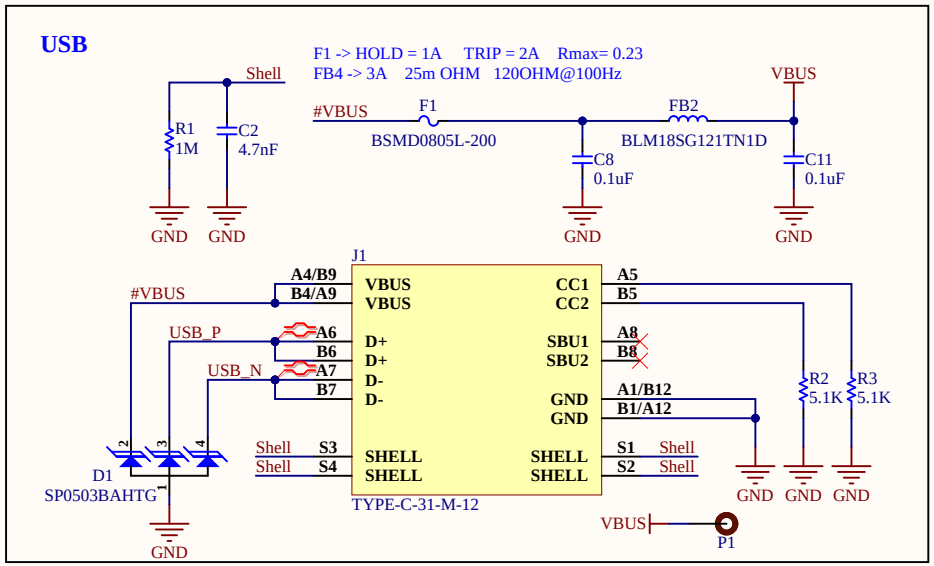


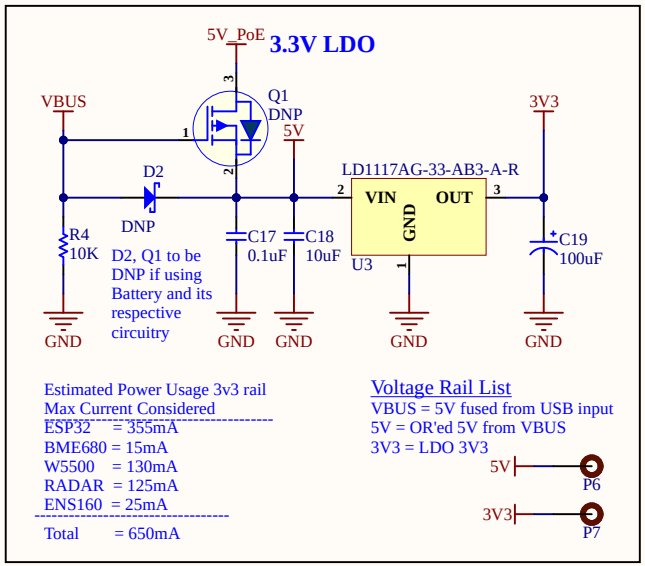
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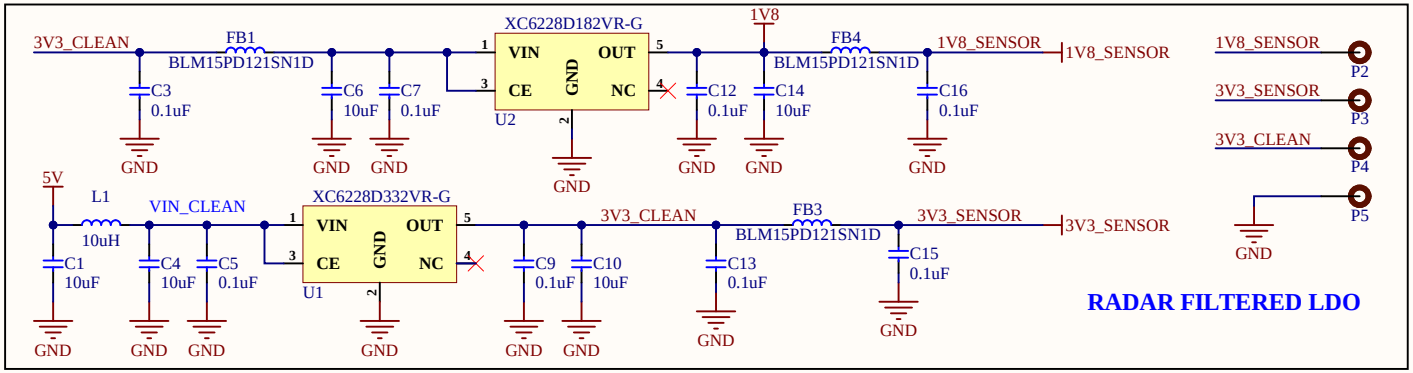
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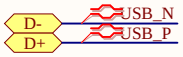
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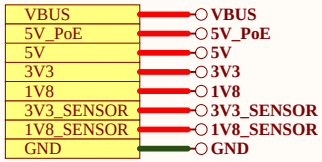
Date: 13/07/2023	Page Num: 1
Drawn By: Huzaifa Ali	Page Total: 8









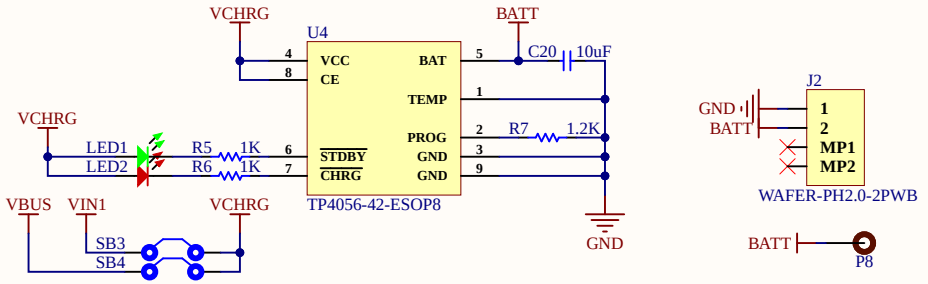


Power Section

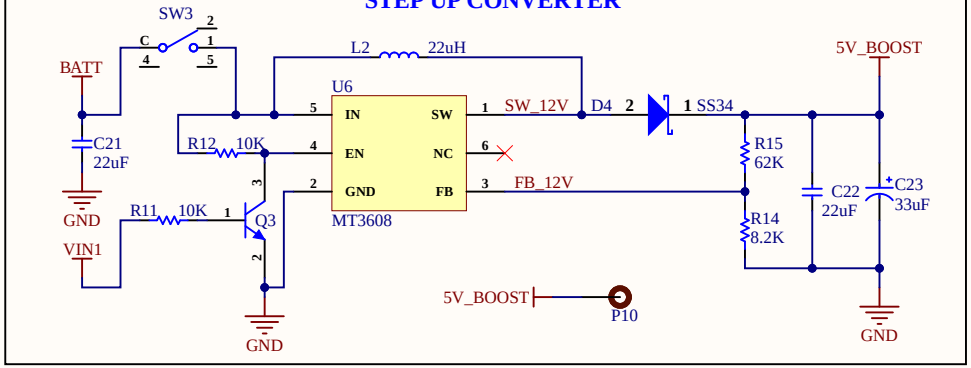
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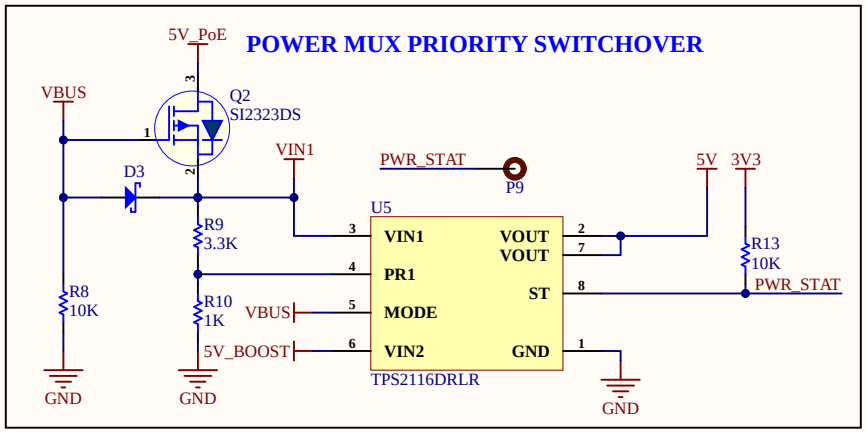
BATTERY CHARGING CIRCUIT



STEP UP CONVERTER



POWER MUX PRIORITY SWITCHOVER



PWR_STAT — PWR_STAT

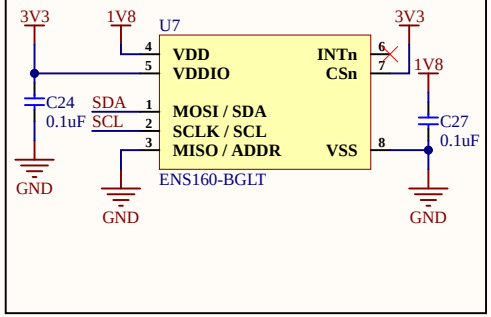
VBUS — VBUS
5V_PoE — 5V_PoE
5V — 5V
3V3 — 3V3
GND — GND

Charging Section

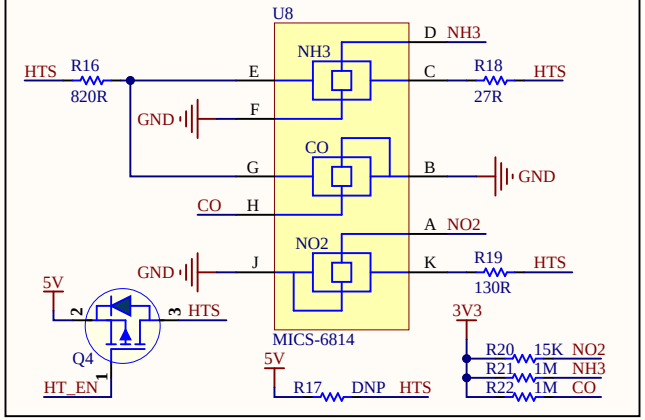
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Title: Charging-Boost-Sheet.SchDoc	
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Date: 13/07/2023	Page Num: 3
Drawn By: Huzaifa Ali	Page Total: 8

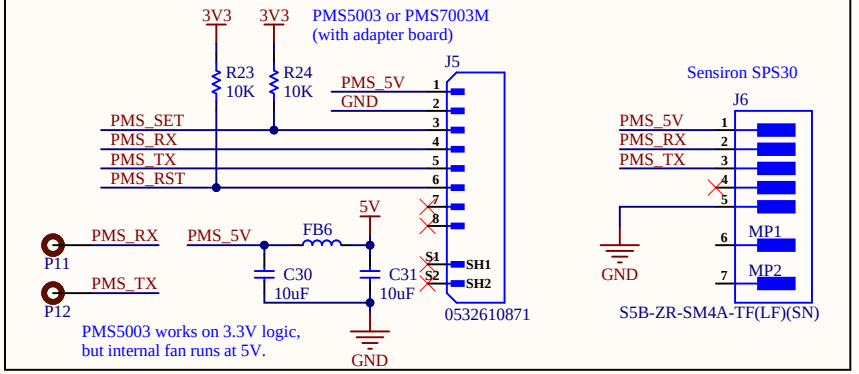
eCO2 TVOC AQI SENSOR



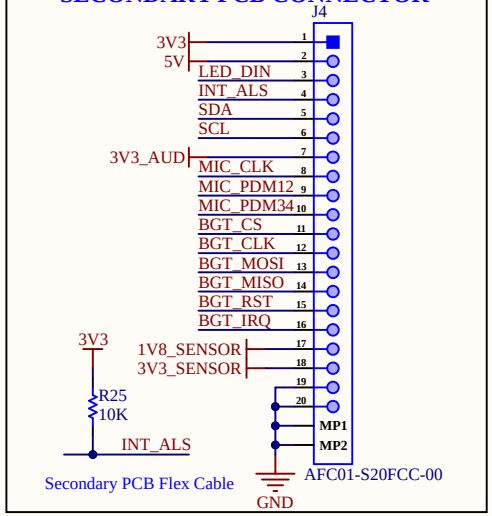
SUPPPORT SENSOR TO DETECT MARIJUANA



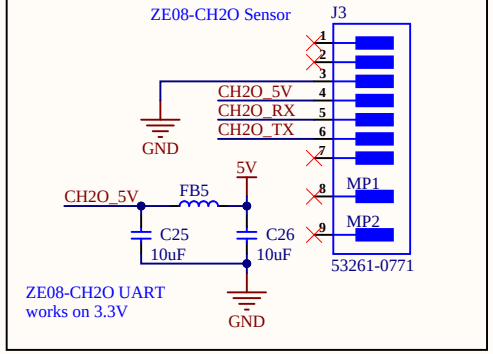
PM2.5 ENVIRONMENT SENSOR CIRCUIT



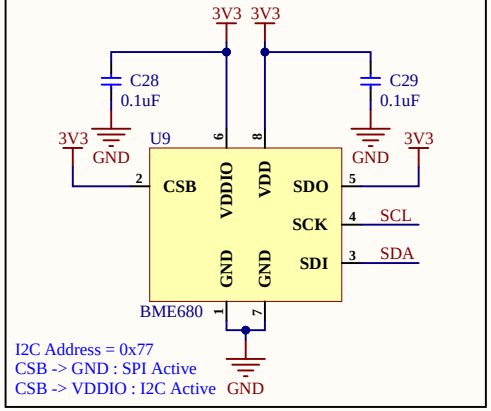
SECONDARY PCB CONNECTOR

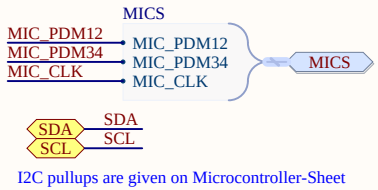


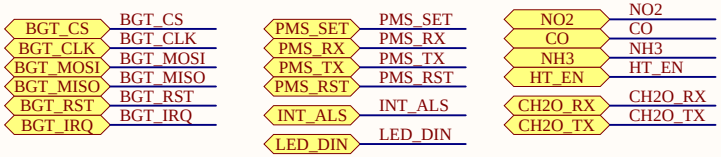
SUPPPORT SENSOR FOR MARIJUANA



TEMP HUM PRESSURE SENSOR





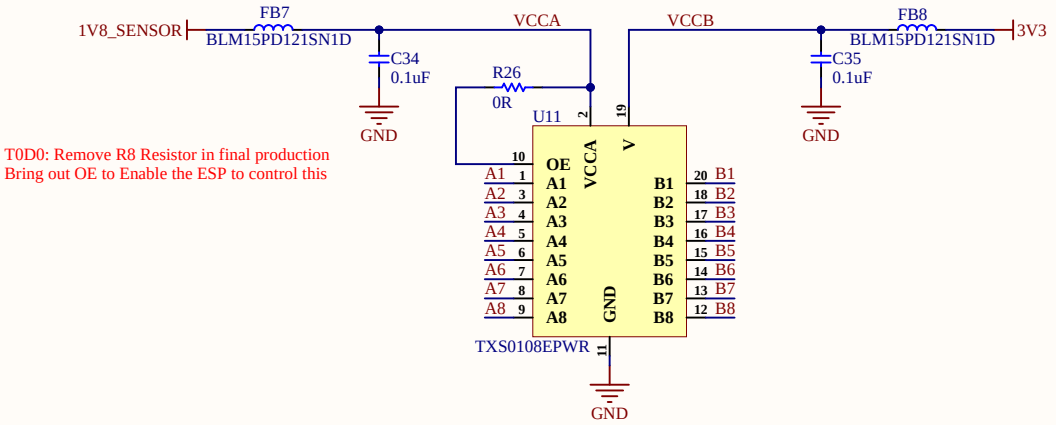


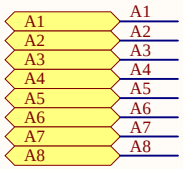
Sensors Section

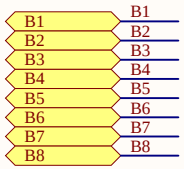
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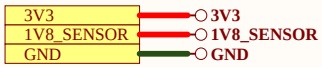
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Date: 13/07/2023	Page Num: 4
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LEVEL SHIFTER





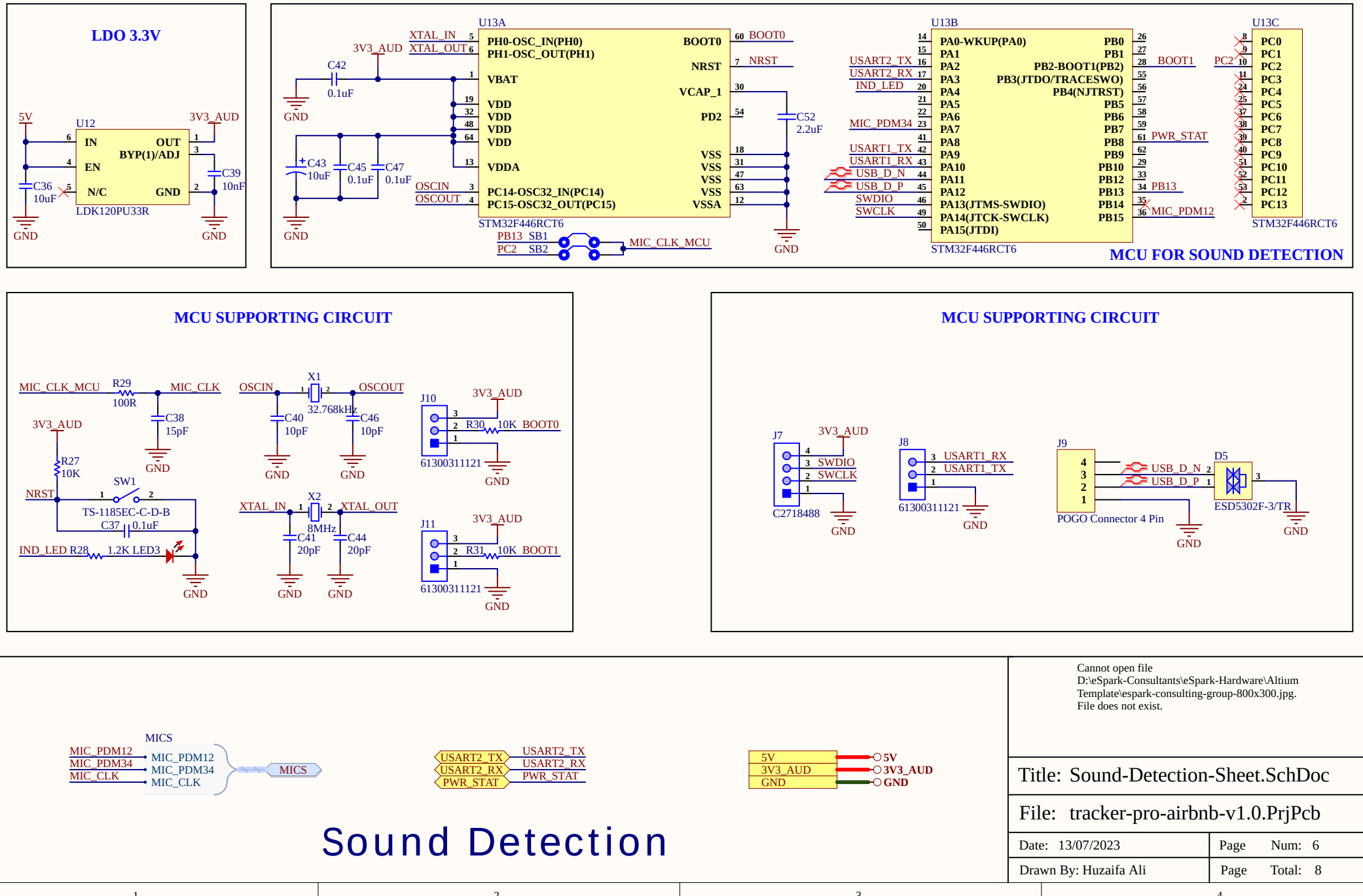


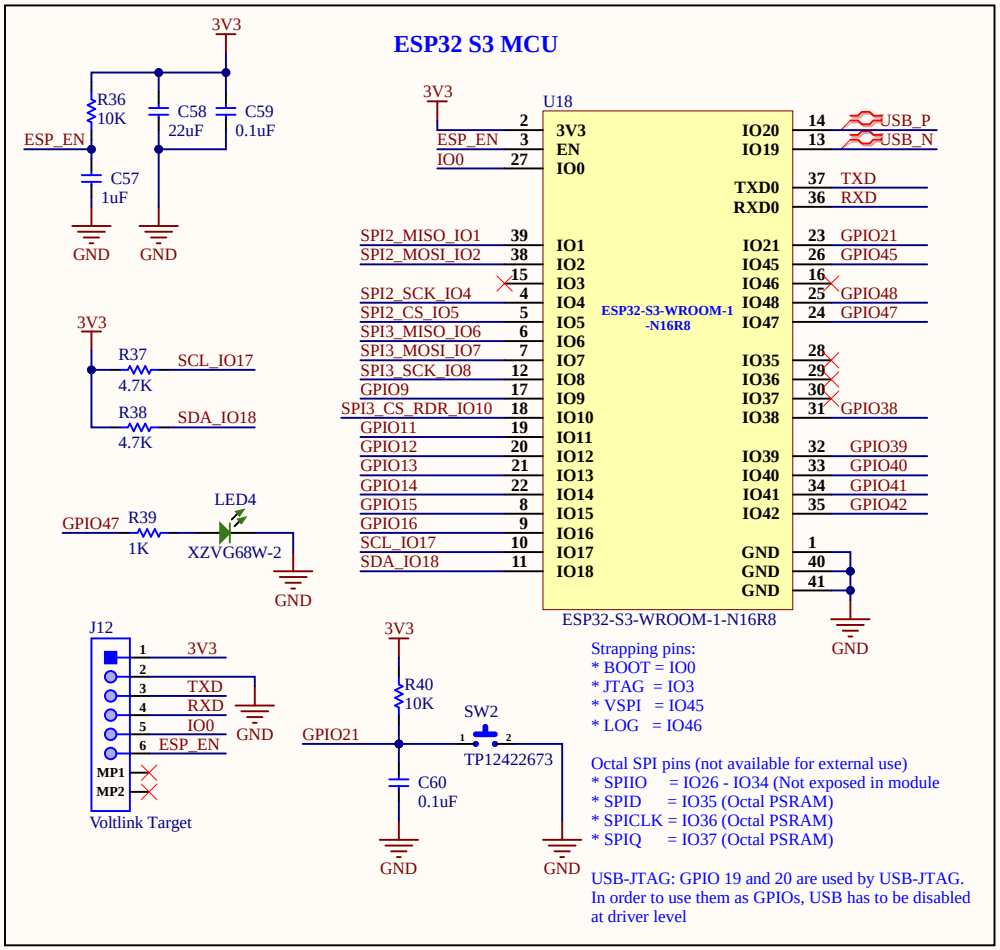


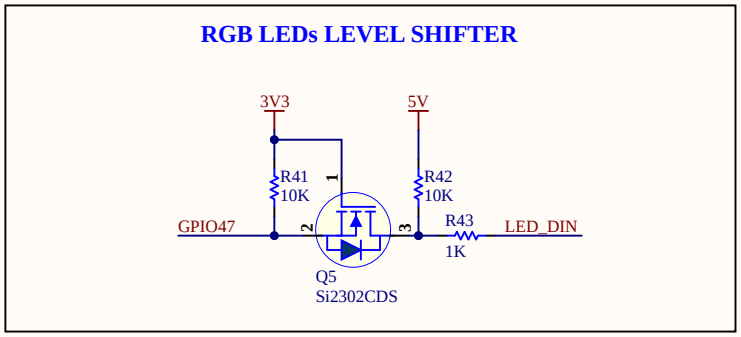
Level Shifter Section

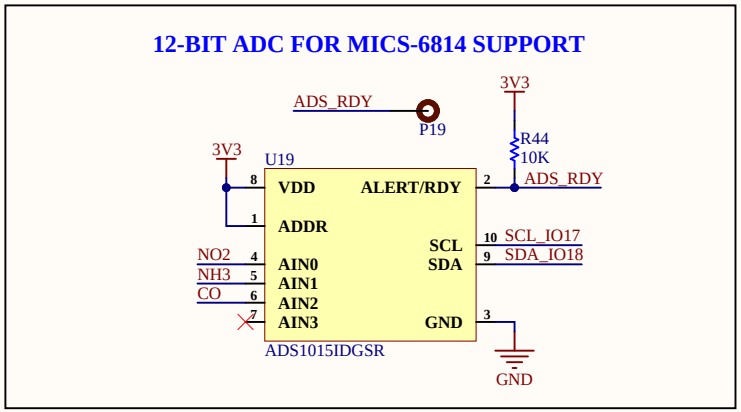
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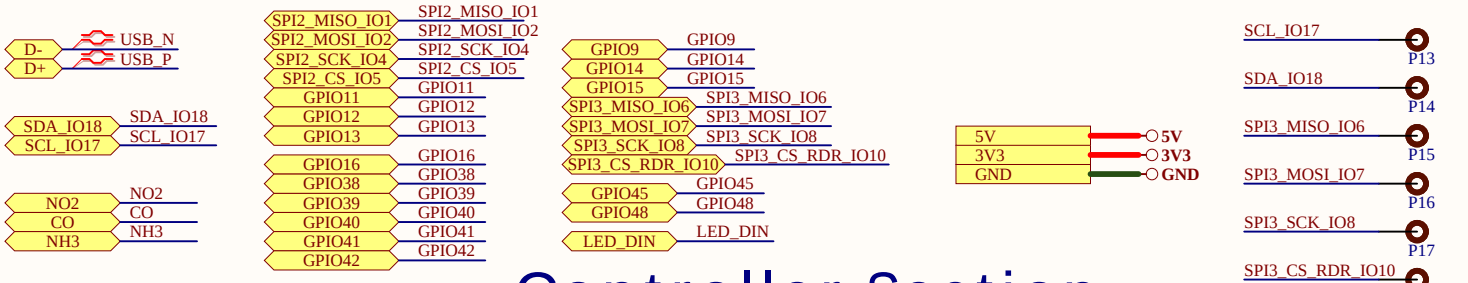
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Date: 13/07/2023	Page Num: 5
Drawn By: Huzaifa Ali	Page Total: 8


LDO 3.3V
5V
U12
IN
OUT
BYP(1)/ADJ
EN
N/C
GND
3V3_AUD
C36
10uF
C39
10nF
LDK120PU33R
GND
GND
U13A
XTAL_IN
XTAL_OUT
3V3_AUD
C42
0.1uF
VBAT
VDD
VDD
VDD
VDD
VDDA
OSCIN
OSCOUT
PC14-OSC32_IN(PC14)
PC15-OSC32_OUT(PC15)
BOOT0
NRST
VCAP_1
PD2
VSS
VSS
VSS
VSS
VSSA
C43
10uF
C45
0.1uF
C47
0.1uF
C52
2.2uF
STM32F446RCT6
PB13 SB1
PC2 SB2
MIC_CLK_MCU
U13B
PA0-WKUP(PA0)
PA1
PA2
PA3
PA4
PA5
PA6
PA7
PA8
PA9
PA10
PA11
PA12
PA13(JTMS-SWDIO)
PA14(JTCK-SWCLK)
PA15(JTDI)
PB0
PB1
PB2-BOOT1(PB2)
PB3(JTDO/TRACESWO)
PB4(NJTRST)
PB5
PB6
PB7
PB8
PB9
PB10
PB12
PB13
PB14
PB15
USART2_TX
USART2_RX
IND_LED
MIC_PDM34
USART1_TX
USART1_RX
USB_D_N
USB_D_P
SWDIO
SWCLK
BOOT1
PWR_STAT
PB13
MIC_PDM12
STM32F446RCT6
U13C
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PC8
PC9
PC10
PC11
PC12
PC13
PC2
STM32F446RCT6
MCU FOR SOUND DETECTION
MCU SUPPORTING CIRCUIT
MIC_CLK_MCU
R29
100R
MIC_CLK
C38
15pF
3V3_AUD
R27
10K
SW1
NRST
TS-1185EC-C-D-B
C37
0.1uF
IND_LED R28
1.2K LED3
GND
OSCIN
X1
32.768kHz
OSCOUT
C40
10pF
C46
10pF
GND
GND
XTAL_IN
X2
8MHz
XTAL_OUT
C41
20pF
C44
20pF
GND
GND
J10
3V3_AUD
R30
10K
BOOT0
61300311121
GND
J11
3V3_AUD
R31
10K
BOOT1
61300311121
GND
MCU SUPPORTING CIRCUIT
J7
3V3_AUD
SWDIO
SWCLK
C2718488
GND
J8
USART1_RX
USART1_TX
61300311121
GND
J9
POGO Connector 4 Pin
USB_D_N
USB_D_P
D5
ESD5302F-3/TR
GND
GND
MICS
MIC_PDM12
MIC_PDM34
MIC_CLK
MIC_PDM12
MIC_PDM34
MIC_CLK
MICS
USART2_TX
USART2_RX
PWR_STAT
USART2_TX
USART2_RX
PWR_STAT
5V
3V3_AUD
GND
5V
3V3_AUD
GND
Sound Detection
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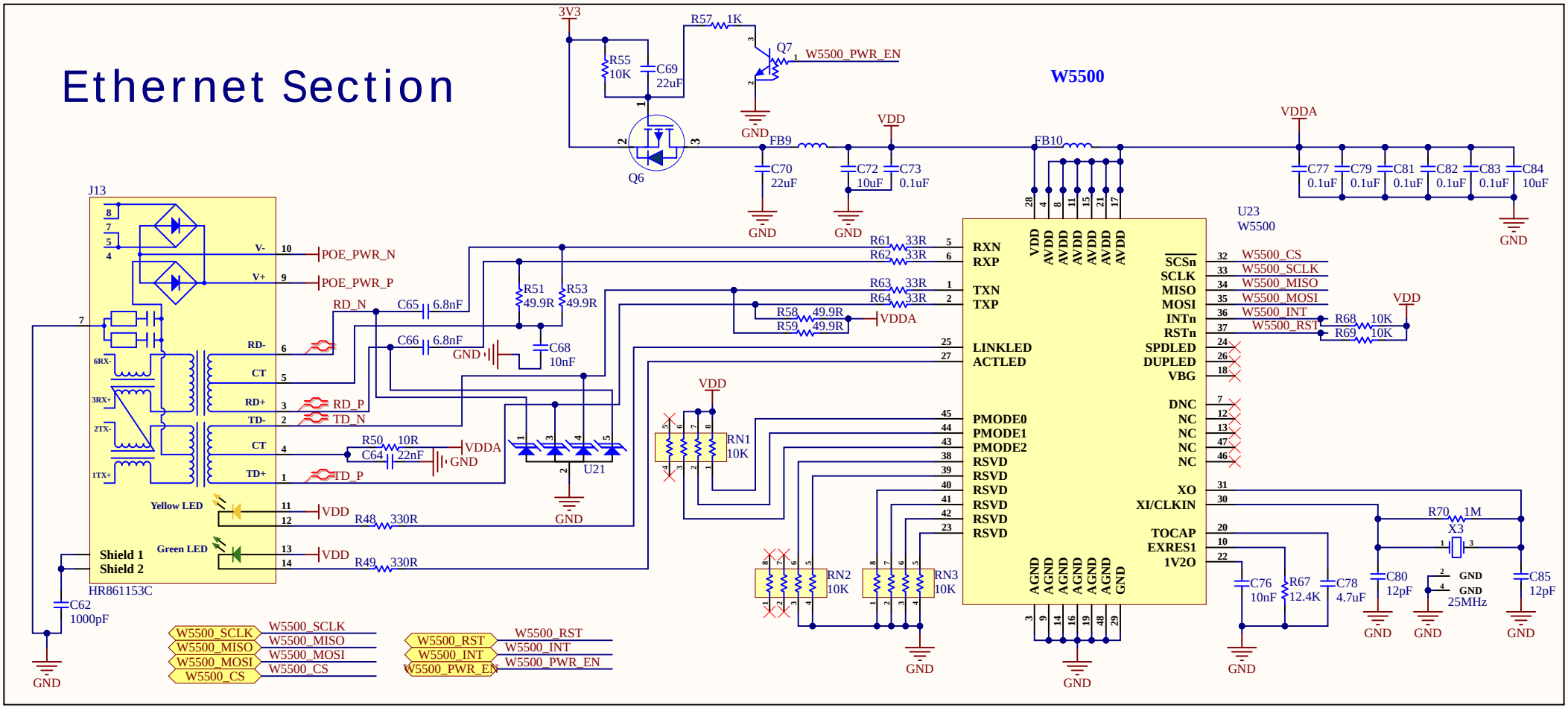
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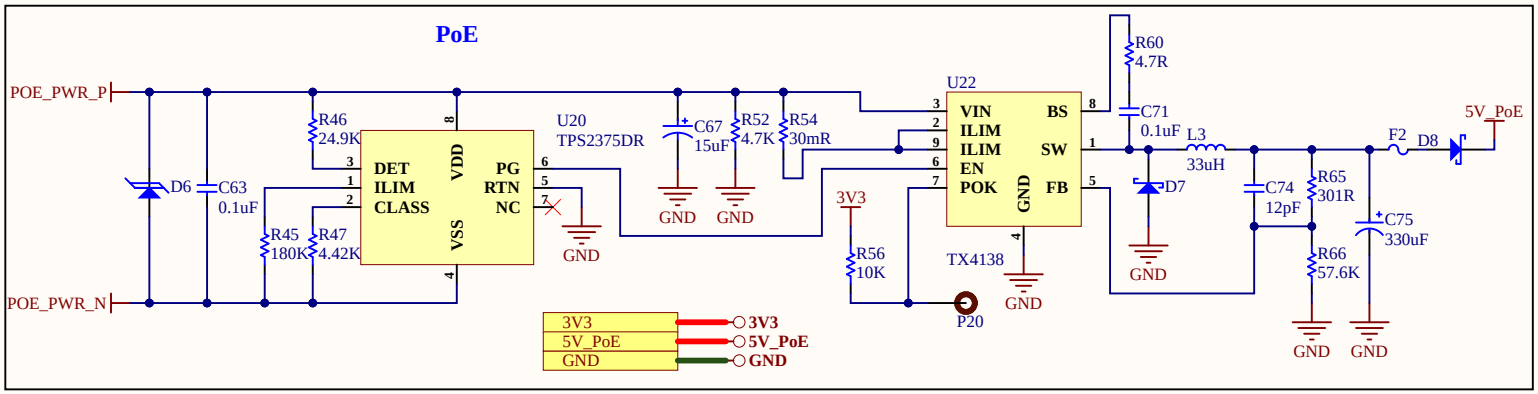
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Ethernet Section



PoE



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